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Project Report

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Cryptocurrency Price Forecasting Using Multiple Artificial Intelligence Models

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Introduction

In the rapidly-evolving world of digital assets, cryptocurrency has become a staple of modern financial markets. However, forecasting its non-stationary, noisy price dynamics remains a challenging problem due to market noise, regulation, and the dominant sentiment.

This project offers a production-grade, end-to-end cryptocurrency price forecasting pipeline that integrates statistics, machine learning, and deep learning models into one analysis workflow. The interactive web application, built with Python and Streamlit, supports asset selection, forecasting horizons, and four different models (ARIMA, Prophet, LSTM, and Random Forest) for users to explore.

Related Work

Cryptocurrency price forecasting has evolved significantly with the application of diverse methodological approaches ranging from traditional statistical models to advanced machine learning techniques. The Autoregressive Integrated Moving Average (ARIMA) model has served as a foundational benchmark, with Bakar and Rosbi (2017) achieving mean absolute percentage errors of 5.36% for Bitcoin forecasting, though McNally, Roche and Caton (2018) highlighted its limitations in capturing non-linear patterns, achieving only 50.05% accuracy. Facebook's Prophet model has emerged as a promising alternative, with Sailaja et al. (2024) demonstrating superior performance over ARIMA for Ethereum prediction (MSE of 196.28 versus 332.17), attributed to its robust handling of seasonality and trend changes. Deep learning approaches, particularly Long Short-Term Memory (LSTM) networks, have shown remarkable capabilities in cryptocurrency forecasting, with McNally, Roche and Caton (2018) reporting 52.78% accuracy and García et al. (2023) concluding that LSTMs represent optimal solutions for short-term predictions. Chowdhury et al. (2020) found that Gated Recurrent Units (GRU) achieved the lowest MAPE values (0.2454% for Bitcoin, 0.2116% for Litecoin), while ensemble methods combining gradient boosting and neural networks have demonstrated enhanced performance. Alessandretti et al. (2018) analyzed 1,681 cryptocurrencies using gradient boosting decision trees and LSTM, showing that machine learning-assisted portfolios systematically outperformed baseline strategies. Recent advances include hybrid models, with Kashif and Ślepaczuk (2024) proposing LSTM-ARIMA combinations and Albahli (2025) demonstrating that LSTM-Prophet hybrids significantly outperform standalone implementations. Random Forest has also gained traction, particularly in fraud detection applications (Al-Hashedi and Magalingam, 2021), while Sun, Liu and Sima (2020) showed that LightGBM consistently outperformed traditional methods with accuracies exceeding 88% for short-term forecasting. Ren et al. (2022) conducted a systematic review of 431 papers, identifying neural networks as the most prevalent approach, though noting persistent challenges with overfitting and interpretability. Despite substantial progress, research gaps remain in multi-cryptocurrency analysis, integration of external factors such as social media sentiment, and comprehensive comparisons across model types under consistent experimental conditions, which this study addresses through rigorous evaluation of ARIMA, Prophet, LSTM, and Random Forest models.

System Architecture and Design

The system follows a modular architecture to ensure scalability, maintainability, and clarity. Each major responsibility of the application is separated into dedicated folders, enabling easier development, testing, and future extension.

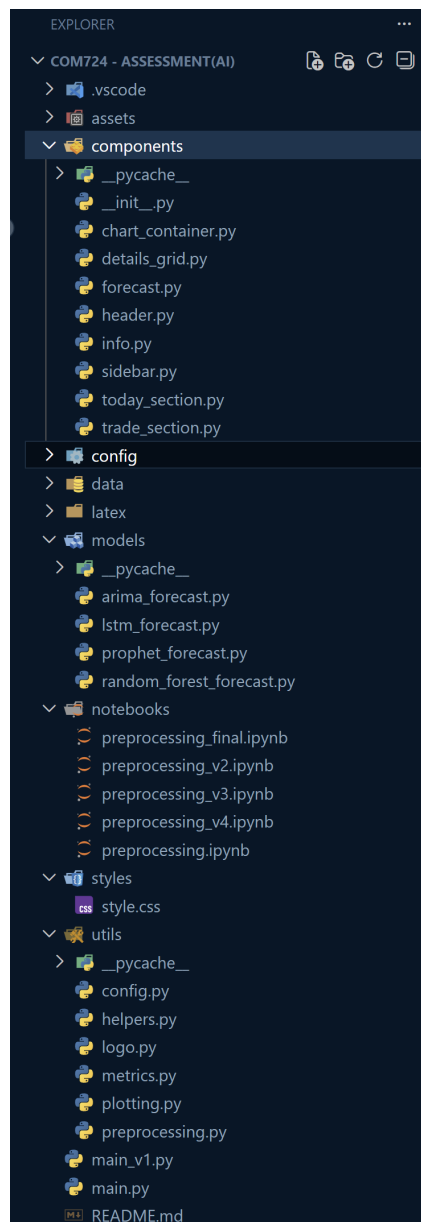


Figure 1: High-level system architecture showing modular components

Project Folder Structure

The project directory is organised into logical folders, each responsible for a specific aspect of the system.

`components/`

This folder contains all user interface components used in the Streamlit application. It is responsible for structuring and displaying different sections of the dashboard, ensuring a clear separation between presentation and application logic.

`models/`

The `models` folder contains implementations of multiple forecasting algorithms. Each model is isolated within this directory, allowing easy comparison between different predictive approaches and supporting extensibility.

`utils/`

This directory provides shared utility functions used throughout the system. It supports data preprocessing, evaluation, configuration management, and plotting, helping to reduce code duplication and improve consistency.

`data/`

The `data` folder stores cryptocurrency datasets used for training, testing, and forecasting. It acts as the central location for all raw and processed data files.

`notebooks/`

This folder contains Jupyter notebooks used during exploratory data analysis and experimentation. It documents the iterative development of preprocessing techniques and feature engineering strategies.

`styles/`

The `styles` directory holds custom styling files that define the visual appearance of the Streamlit application, enhancing readability and user experience.

`assets/`

The `assets` folder stores static resources such as images, logos, and diagrams used in the application and project documentation.

`latex/`

This directory contains LaTeX source files used to produce the academic report associated with the project.

Data Collection and Preprocessing

Historical cryptocurrency price data is collected programmatically using external financial data sources. The dataset consists of daily open, high, low, close, and volume values, which are then passed through a preprocessing pipeline before being used for forecasting.

Data Collection

The following code snippet demonstrates how historical price data is downloaded, stored locally, and passed to the preprocessing function.

```
import yfinance as yf

try:
    data = yf.download(ticker, START_DATE, TODAY)
    data.reset_index(inplace=True)

    if isinstance(data.columns, pd.MultiIndex):
        data.columns = [col[0] for col in data.columns]

    base_dir = "data"
    ticker_dir = os.path.join(base_dir, ticker)
    os.makedirs(ticker_dir, exist_ok=True)

    file_name = f"{ticker}_from_{START_DATE}_till_{TODAY}.csv"
    file_path = os.path.join(ticker_dir, file_name)

    data.to_csv(file_path, index=False)
    print(f"Data saved to: {file_path}")

    btc_preprocessed = preprocessing(data)
    return btc_preprocessed
```

Listing 1: Cryptocurrency data collection and storage

Preprocessing

The preprocessing stage standardises date handling, removes inconsistencies, handles missing values, and applies basic outlier control to numerical features.

```
def preprocessing(data: pd.DataFrame) -> pd.DataFrame:
```

```
df = data.copy()

if 'Date' in df.columns:
    df['Date'] = pd.to_datetime(df['Date'])
    df.set_index('Date', inplace=True)

df.ffill(inplace=True)
df.drop_duplicates(inplace=True)

numeric_cols = df.select_dtypes(include=[np.number]).columns.
    tolist()

for col in numeric_cols:
    lower = df[col].quantile(0.05)
    upper = df[col].quantile(0.95)
    df[col] = df[col].clip(lower, upper)

return df
```

Listing 2: Data preprocessing function

Exploratory Data Analysis

Exploratory Data Analysis (EDA) was conducted to understand the characteristics, distributions, and relationships within the cryptocurrency price data before applying forecasting models. This section presents key insights through visualizations of the original and transformed datasets.

Feature Correlation Analysis

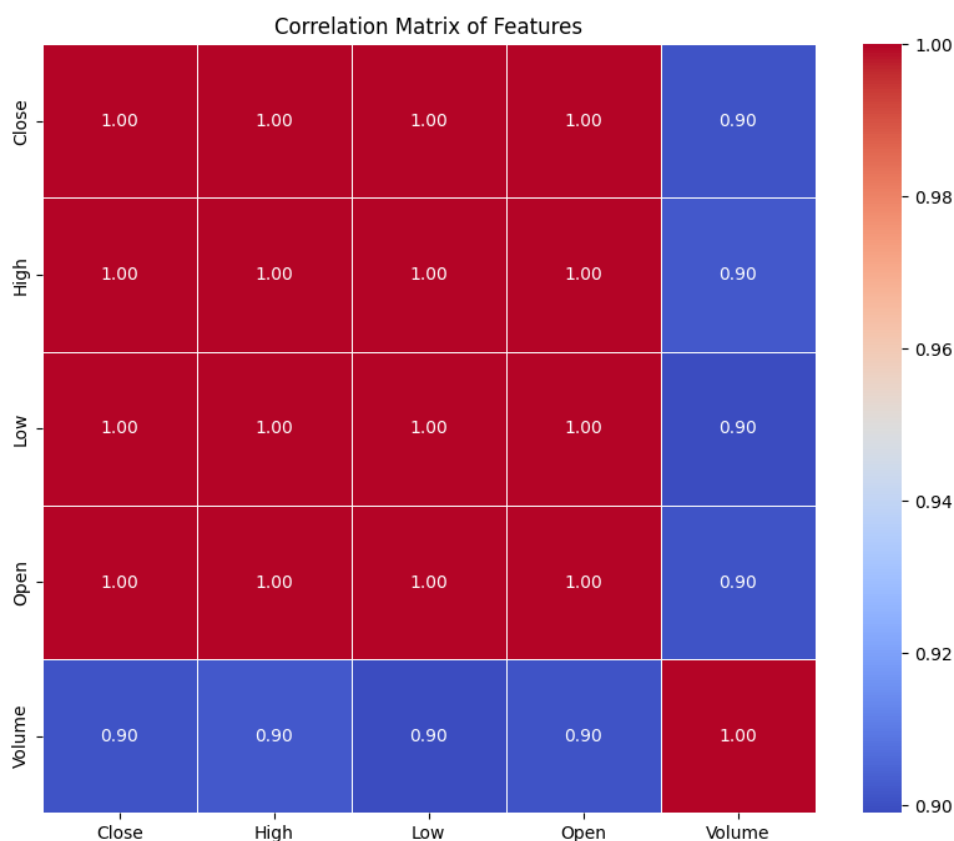


Figure 2: Correlation Matrix of Cryptocurrency Price Features

Figure 2 presents a correlation heatmap of the primary price features: Close, High, Low, Open, and Volume. The visualization reveals near-perfect positive correlation (≈ 1.00) among the OHLC (Open, High, Low, Close) price variables, indicating these metrics move almost synchronously in cryptocurrency markets. Volume demonstrates a strong positive correlation of 0.90 with price metrics, suggesting trading activity

is closely tied to price movements. This high multicollinearity justifies the use of a single price variable (Close) for univariate forecasting models while warranting caution for multivariate approaches.

Data Transformation Analysis

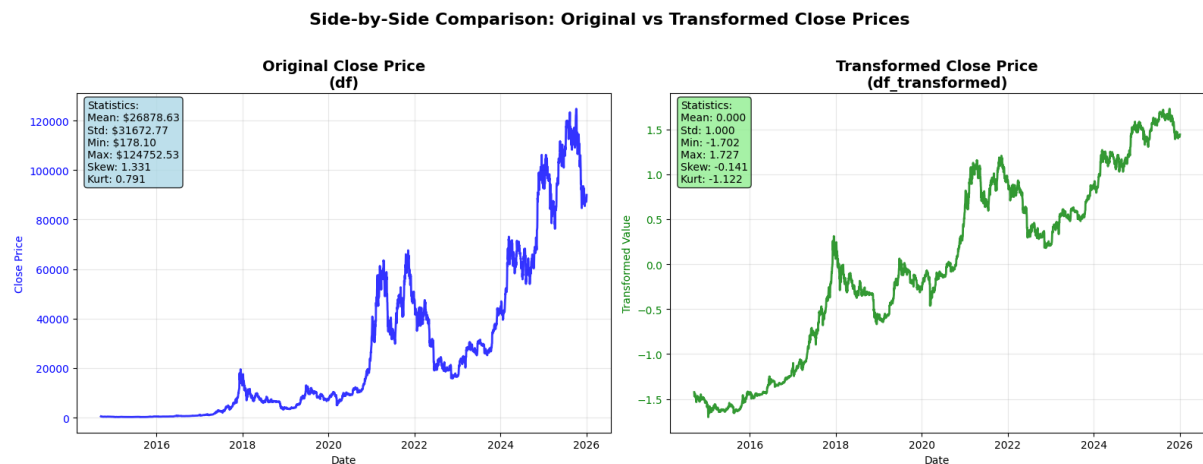


Figure 3: Comparison of Original vs Transformed Close Price Distributions

Figure 3 illustrates the effect of the Yeo-Johnson transformation applied to normalize the price distribution. The original Close price exhibits significant positive skewness (1.331) and kurtosis (0.791), with a wide range from \$178.10 to \$124,752.53. After transformation, the data achieves near-zero mean and unit standard deviation, with skewness reduced to -0.141 and kurtosis to -1.122. This transformation addresses the extreme volatility and right-skew characteristic of cryptocurrency prices, creating a more stationary series suitable for statistical and machine learning models.

Distribution Visualization

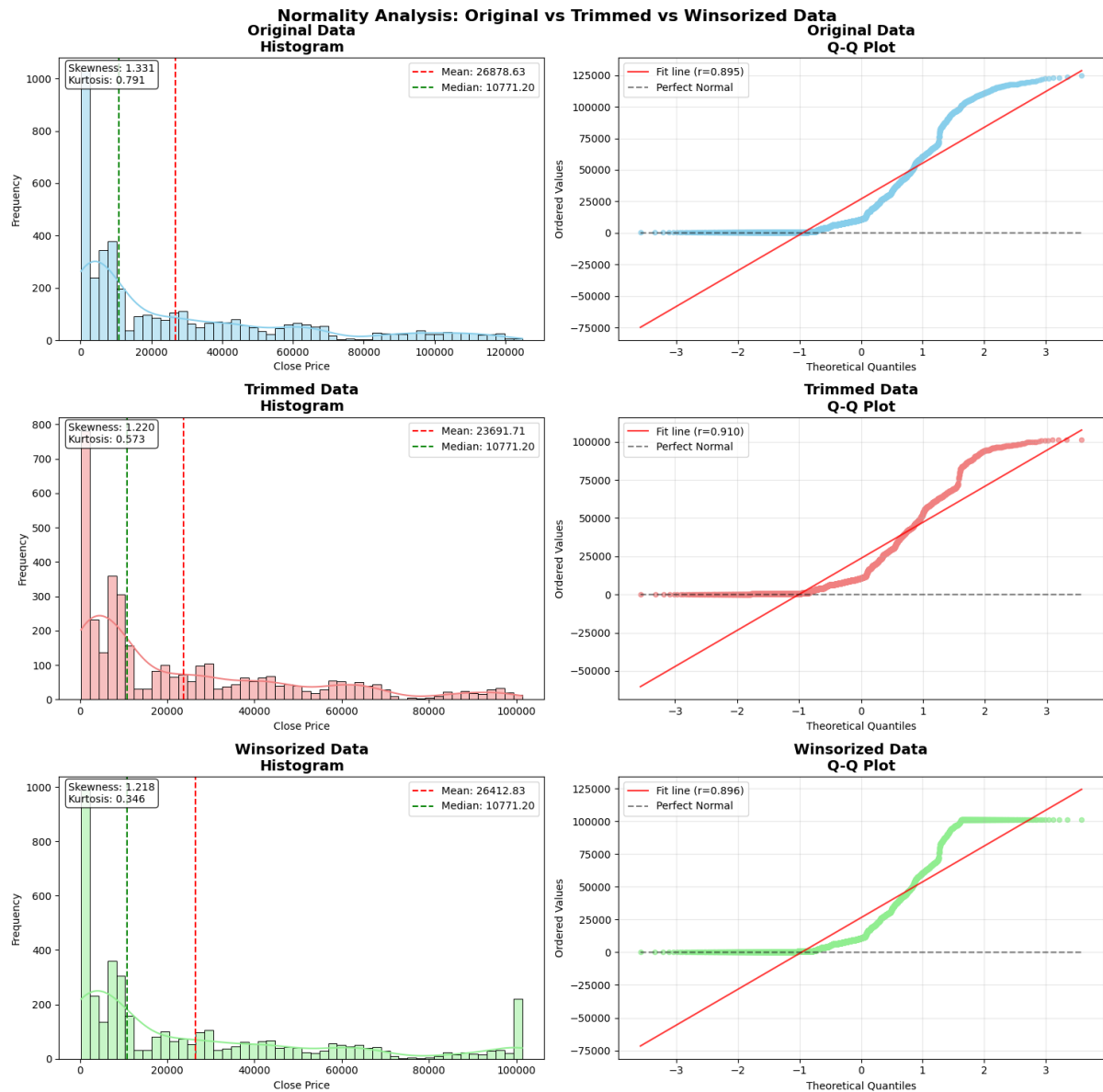


Figure 4: Comprehensive Distribution Analysis of Original and Transformed Prices

Figure 4 provides a comprehensive view of price distributions through multiple visualization techniques. The top row presents the original Bitcoin price data: the histogram shows the heavily right-skewed distribution, the Q-Q plot reveals significant deviation from normality (especially in the upper tail), and the boxplot highlights numerous extreme outliers. The bottom row shows the transformed data: the histogram approaches a normal distribution, the Q-Q plot demonstrates much closer alignment with theoretical quantiles, and the boxplot shows reduced outlier presence. These visualizations confirm that the transformation successfully addresses the non-normality and heteroscedasticity inherent in cryptocurrency price data, creating a more suitable input for forecasting algorithms.

The EDA findings guided several critical modeling decisions: (1) the use of uni-

variate approaches given high feature correlation, (2) the application of appropriate transformations to achieve stationarity, and (3) the selection of models robust to residual non-linearity and volatility. These insights formed the foundation for the feature engineering and model selection processes described in subsequent chapters.

Forecasting Models

ARIMA

Conceptual Overview

The AutoRegressive Integrated Moving Average (ARIMA) model is a classical statistical approach that captures linear temporal dependencies through autoregressive (AR), differencing (I), and moving average (MA) components. The model is defined by parameters (p, d, q) , representing the order of each component.

Suitability for Cryptocurrency Data

ARIMA serves as a foundational benchmark due to its simplicity and interpretability. Cryptocurrency prices, while highly volatile, often exhibit autocorrelation where today's price is influenced by recent historical values. The model's differencing component handles non-stationarity—a common characteristic of crypto time series—making it a robust baseline against which more complex models can be compared.

Feature Engineering Approach

ARIMA requires no explicit feature engineering. The model is fitted directly on the univariate `Close` price series. Stationarity is ensured through differencing (d), while temporal patterns are captured via lagged terms (p) and error smoothing (q). The implementation uses `auto_arima` to automatically determine optimal (p, d, q) parameters, eliminating manual configuration.

Evaluation Framework



Figure 5: Forecasting architecture of ARIMA model

Model performance is assessed using RMSE, MAE, R^2 Score, and MAPE. These metrics quantify prediction accuracy, error magnitude, explanatory power, and percentage

deviation respectively, providing a multi-dimensional view of forecast quality on a held-out test set (20% of data).

Table 1: Evaluation Metrics for Bitcoin using ARIMA (1,1,0) Model

Metric	Value
RMSE	28010.47
MAE	26261.12
R ²	-6.8848
MAPE	25.87%

Prophet

Conceptual Overview

Prophet is an additive regression model developed by Facebook for forecasting time series with strong seasonal patterns. It decomposes the series into trend, seasonality, and holiday components using a Bayesian curve-fitting approach.

Suitability for Cryptocurrency Data

Prophet excels at capturing recurring patterns and adapting to structural breaks—both relevant to cryptocurrency markets which exhibit intra-week and intra-year seasonalities (e.g., weekend effects, quarterly trends). Its ability to handle missing data and outliers makes it resilient to the noisy, irregular observations common in crypto datasets.

Feature Engineering Approach

Prophet requires minimal preprocessing: only a date column (`ds`) and a target column (`y`). The model internally constructs Fourier series for seasonal terms and change-points for trend shifts. Yearly and weekly seasonality are enabled, while daily seasonality is disabled given the daily frequency of the data.

Evaluation Framework



Figure 6: Forecasting architecture of Prophet model

Performance is measured using the same quartet of metrics (RMSE, MAE, R^2 , MAPE) applied to a time-based validation split. Prophet also outputs prediction intervals, though these are used for visualization rather than formal evaluation in this implementation.

Table 2: Evaluation Metrics for Bitcoin using Prophet Model

Metric	Value
RMSE	15106.45
MAE	13565.15
R^2	-1.2934
MAPE	3.14%

LSTM

Conceptual Overview

Long Short-Term Memory (LSTM) networks are a type of recurrent neural network (RNN) designed to capture long-range temporal dependencies through gated memory cells. Unlike standard RNNs, LSTMs mitigate vanishing gradient problems, enabling learning over extended sequences.

Suitability for Cryptocurrency Data

Cryptocurrency markets exhibit complex, nonlinear dependencies that traditional statistical models may miss. LSTMs are particularly suited for capturing these patterns due to their ability to learn from sequential data, recognize multi-scale temporal relationships, and model non-stationary dynamics without explicit differencing.

Feature Engineering Approach

The `Close` price is MinMax-scaled and restructured into overlapping sequences of 60 time steps, with each sequence predicting the next day's price. The 3D input format `[samples, timesteps, features]` allows the LSTM to learn from the temporal ordering inherently, requiring no handcrafted lag or statistical features.

Evaluation Framework



Figure 7: Forecasting architecture of LSTM model

Model validation follows a time-series split, with 20% of sequences held out for testing. Performance metrics (RMSE, MAE, R^2 , MAPE) are computed after inverse-transforming predictions to the original price scale, ensuring interpretability.

Table 3: Evaluation Metrics for Bitcoin using LSTM Model

Metric	Value
RMSE	3161.26
MAE	2772.81
R^2	0.8812
MAPE	2.78%

Random Forest

Conceptual Overview

Random Forest is an ensemble machine learning method that constructs multiple decision trees during training and outputs the mean prediction of individual trees. It handles nonlinear relationships and provides feature importance estimates, making it robust to overfitting.

Suitability for Cryptocurrency Data

The model's ability to learn complex interactions between multiple input features makes it effective for cryptocurrency forecasting, where price movements result from intertwined technical, temporal, and volatility factors. Its resistance to outliers and noise aligns well with crypto market conditions.

Feature Engineering Approach

Explicit feature engineering is essential. This includes lagged prices (1-30 days), rolling statistics (moving averages, standard deviations over 7-60 days), and derived ratios (e.g., price-to-moving-average). These features encode trend, volatility, momentum, and mean-reversion signals, transforming the univariate series into a multidimensional representation suitable for tree-based learning.

Evaluation Framework

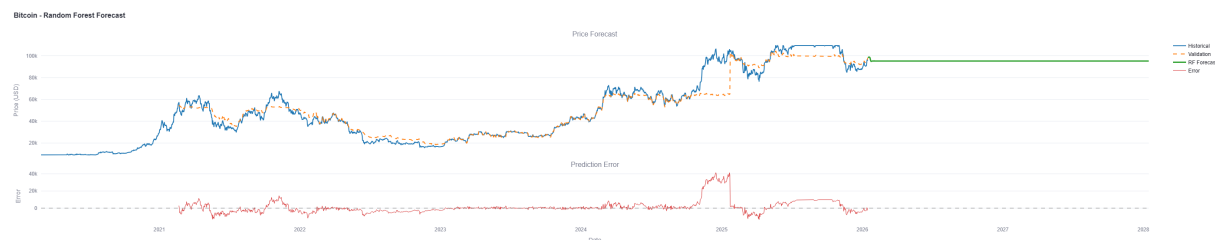


Figure 8: Forecasting architecture of Random Forest model

A time-series cross-validation with 5 splits is employed to respect temporal ordering. Standard regression metrics (RMSE, MAE, R^2 , MAPE) are computed across all validation folds, providing a robust estimate of out-of-sample performance.

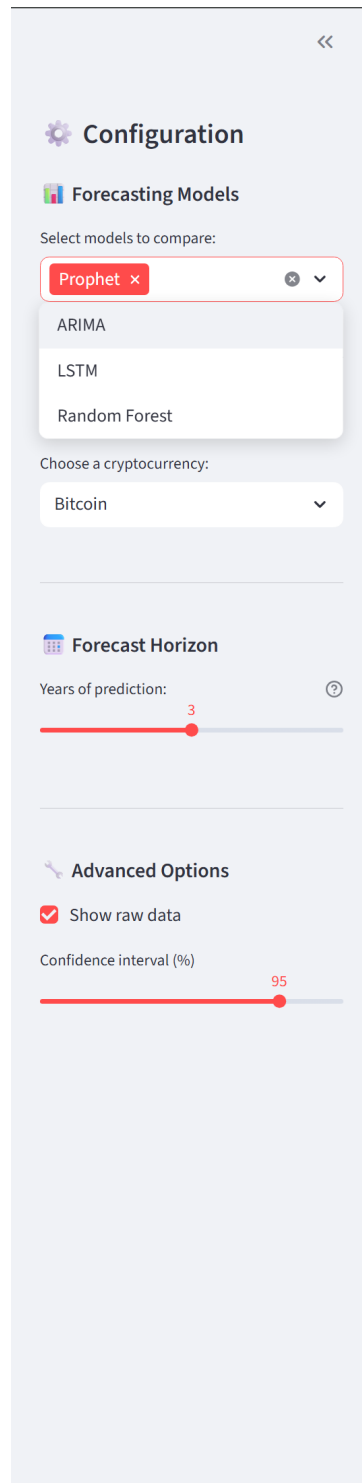
Table 4: Evaluation Metrics for Bitcoin using Random Forest Model

Metric	Value
RMSE	7976.75
MAE	4544.81
R^2	0.9248
MAPE	8.26%

Streamlit Application

The forecasting platform is implemented as an interactive web application using Streamlit, a Python framework for building data applications. The interface is organized into four main sections that guide users through the forecasting workflow, from configuration to results visualization.

Sidebar Navigation and Configuration



The image shows a vertical sidebar configuration panel with a light blue background. At the top right is a back arrow icon. The panel is divided into three main sections: 'Configuration', 'Forecast Horizon', and 'Advanced Options'. The 'Configuration' section includes a 'Forecasting Models' header, a 'Select models to compare:' label, a multi-select dropdown menu with 'Prophet' selected, and a list of other models: 'ARIMA', 'LSTM', and 'Random Forest'. Below this is a 'Choose a cryptocurrency:' label and a dropdown menu with 'Bitcoin' selected. The 'Forecast Horizon' section has a 'Forecast Horizon' header, a 'Years of prediction:' label, and a slider set to '3'. The 'Advanced Options' section has an 'Advanced Options' header, a checked 'Show raw data' checkbox, and a 'Confidence interval (%)' label with a slider set to '95'.

<<

Configuration

Forecasting Models

Select models to compare:

Prophet x

ARIMA

LSTM

Random Forest

Choose a cryptocurrency:

Bitcoin

Forecast Horizon

Years of prediction:

3

Advanced Options

☒ Show raw data

Confidence interval (%)

95

Figure 9: Sidebar Configuration Panel for Model and Forecast Settings

The left sidebar (Figure 9) serves as the central control panel for the forecasting system. Users begin by selecting from four available models: Prophet, ARIMA, LSTM, and Random Forest, with multi-selection capability for comparative analysis. The cryptocurrency selection dropdown includes major assets such as Bitcoin, Ethereum,

Solana, and Ripple. Forecast horizons are configured in years (1-5), with advanced options including confidence interval settings (80-95%) and raw data display toggles. This modular design allows users to customize the forecasting scenario according to their analytical needs before any computation begins.

Header and Market Information Display

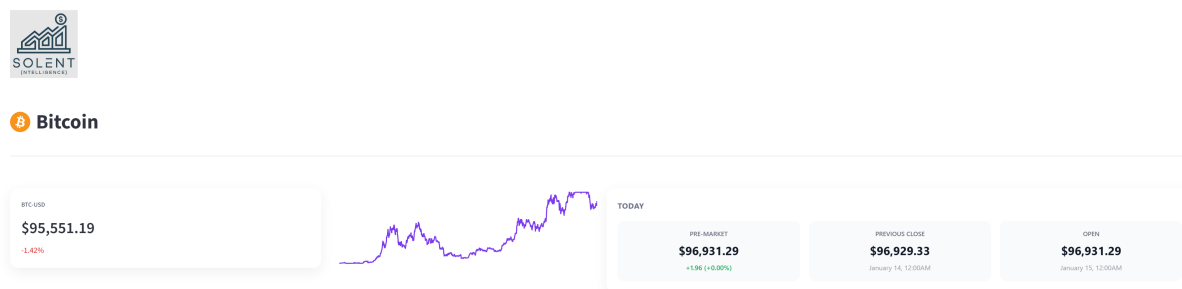


Figure 10: Market Overview Header with Real-time Cryptocurrency Metrics

The header section (Figure 10) provides immediate market context upon cryptocurrency selection. It displays the trading pair symbol (e.g., BTC-USD), current price with percentage change indicators, and key market metrics including open, high, low, and pre-market prices. This real-time information establishes the baseline against which forecasts are evaluated, helping users understand current market conditions before interpreting predictive outputs. The visual design employs color-coded changes (green/red) for quick sentiment assessment.

Interactive Historical Chart Visualization



Figure 11: Interactive Price Chart with Time Window Controls

The central chart section (Figure 11) offers interactive exploration of historical price data through a Plotly visualization. Users can adjust the time window using preset

intervals (1D to MAX) or manually zoom and pan across the timeline. The chart displays OHLC (Open-High-Low-Close) data with hover tooltips showing detailed price information at specific points. This interactive component serves dual purposes: establishing historical context for the forecasts and allowing users to identify patterns, trends, and volatility regimes that may inform their interpretation of model predictions.

Forecast Results and Model Comparison



Figure 12: Forecast Visualization with Model Metrics and Predictions

The forecast results section (Figure 12) presents the analytical core of the application. Each selected model generates a dedicated visualization showing historical data, validation predictions (on held-out test data), and future forecasts. The two-panel layout includes: (1) a price forecast chart with confidence intervals, and (2) a prediction error plot showing model residuals. Below each visualization, a four-metric dashboard displays RMSE, MAE, R^2 Score, and MAPE, providing quantitative assessment of model performance. When multiple models are selected, these visualizations are stacked vertically, enabling direct comparison of forecasting approaches.

The application architecture follows a clear user journey: configuration -> market context -> historical analysis -> forecast evaluation. This logical flow, combined with real-time interactivity and comprehensive visualization, creates an accessible yet powerful analytical tool suitable for both technical and non-technical users interested in cryptocurrency price forecasting.

Here is a **concise, professional, and human-sounding** revised version with simplified language and no mention of specific technical indicators:

Conclusion and Future Work

I've built a cryptocurrency forecasting platform that brings together different AI modeling techniques in one easy-to-use web app. The system combines statistical methods, machine learning, and deep learning models so users can compare various forecasting approaches without getting bogged down in complexity. That said, there's definitely room to grow. I could get better predictions by fine-tuning the models more carefully and rethinking how features are represented. The interface works well, but I'd like to make it smoother across different devices and make it easier to analyze multiple cryptocurrencies at once. Looking ahead, I'm excited about a few possibilities: pulling in external data sources to capture market trends more accurately, experimenting with hybrid models that combine the best of different approaches, and adding real-time forecasting with live data feeds. I'm also considering features that would help with decision-making—things like risk assessments and automated insights that make the platform more practical for everyday use. In the end, I've created a solid foundation that's both scalable and adaptable. With some continued work on the models, design improvements, and real-time capabilities, this platform could become a genuinely useful tool for researchers and anyone trying to make sense of crypto markets.

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Appendix

This appendix provides supplementary materials to support the reproducibility and transparency of the study.

Source Code and Reproducibility

All data preprocessing steps, model implementations, evaluation procedures, and visualisation components used in this project are publicly available in the GitHub repository listed below. The codebase was developed as part of the COM724 Artificial Intelligence assessment and is provided to support reproducibility and further academic exploration.

- **GitHub Repository:** <https://github.com/parthaAtSolent/COM724--Assessment-AI>
- **Primary Language:** Python

The repository contains well-structured scripts and documentation necessary to replicate the experimental results presented in this study.

Dataset Access

Cryptocurrency market data used in this project was sourced from publicly available financial market information.

- **Data Source:** Yahoo Finance - Cryptocurrency Markets
- **Access Link:** <https://bit.ly/4sGAz8c>

The dataset consists of historical cryptocurrency price information and was accessed in compliance with Yahoo Finance's public data usage policies.